AttenX: Overcoming Global Incoherence in Text-to-Image Synthesis via Self-Attention and Spectral Normalization

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***Abstract*—Synthesizing high-fidelity, photorealistic images from natural language descriptions is a persistent challenge in multimodal machine learning. While multi-stage architectures like the Attentional Generative Adversarial Network (AttnGAN) have significantly improved fine-grained synthesis using word-level attention, they suffer from critical drawbacks. Specifically, AttnGAN often fails to capture long-range spatial dependencies, resulting in anatomically deformed objects and a lack of global structural coherence. Furthermore, the reliance on multiple discriminators across different resolutions induces severe training instability and mode collapse. In this paper, we propose AttenX, a novel generative architecture designed to address these limitations. AttenX integrates a Non-Local Self-Attention module within the generator's intermediate stages to enforce global spatial consistency, ensuring that anatomically distant features remain coherent. To mitigate training instability, we apply Spectral Normalization across all multi-scale discriminators, constraining their Lipschitz continuity and enabling the use of a Two Time-Scale Update Rule (TTUR). Experimental simulations on the CUB-200-2011 dataset demonstrate that AttenX yields a 12.16% improvement in the Inception Score (IS) and a 22.64% reduction in Fréchet Inception Distance (FID) compared to the baseline AttnGAN. Qualitative results confirm a substantial reduction in morphological hallucinations, paving the way for more robust and anatomically accurate text-to-image synthesis.**

***Index Terms*—Generative Adversarial Networks, Text-to-Image Synthesis, Self-Attention, Spectral Normalization, AttnGAN, Deep Learning.**

I. Introduction

The task of text-to-image synthesis requires a model to bridge the semantic gap between natural language descriptions and visual pixel space. Early approaches, such as GAN-INT-CLS [5], demonstrated the viability of conditional GANs but were limited to low-resolution outputs. Innovations like StackGAN [6] introduced multi-stage refinement to reach 256×256, yet lacked fine-grained control over localized visual attributes.

The Attentional Generative Adversarial Network (AttnGAN) [1] revolutionized the field by introducing the Deep Attentional Multimodal Similarity Model (DAMSM), allowing the generator to synthesize specific sub-regions based on individual words. However, as shown in Fig. 1(a), AttnGAN's reliance on local convolutions often leads to "Frankenstein" artifacts where anatomical parts are disconnected.

A. Drawbacks of the Baseline Model

Analysis reveals two critical flaws: (1) **Lack of Global Coherence** due to the local receptive fields of standard CNNs, and (2) **Training Instability** as multi-scale discriminators converge too quickly, causing mode collapse.

II. Related Work

Reed et al. [5] established the foundation for mapping text to visual distributions. Zhang et al. [6], [7] evolved this into multi-stage synthesis. AttnGAN [1] further refined this with word-level attention. More recently, SAGAN [2] demonstrated that self-attention can model long-range dependencies in unconditional generation. AttenX adapts these concepts to the conditional text-to-image domain while stabilizing the adversarial collision via Spectral Normalization [3].

III. Proposed Methodology: AttenX

AttenX retains the F_0, F_1, F_2 hierarchical structure but injects a Non-Local Self-Attention module and constrains the discriminators mathematically.

A. Global Coherence via Self-Attention

To introduce global awareness, we design a Self-Attention module that calculates the response at a position as a weighted sum of features at all spatial locations. Given feature map $X \in \mathbb{R}^{C \times N}$, we project it into Query (Q), Key (K), and Value (V) spaces:

$$Q = W_q X, \quad K = W_k X, \quad V = W_v X \tag{1}$$

The attention map A is computed as:

$$A_{j,i} = \frac{\exp(Q_i^T K_j)}{\sum_{i=1}^{N} \exp(Q_i^T K_j)} \tag{2}$$

The final output is a residual connection: $Y_i = \gamma O_i + X_i$, where γ is a learnable parameter initialized to zero.

B. Adversarial Stabilization

We apply Spectral Normalization (SN) [3] to all convolutional weights W in discriminators D_0, D_1, D_2:

$$\bar{W}_{SN} = \frac{W}{\sigma(W)} \tag{3}$$

where $\sigma(W)$ is the spectral norm. This bounds the Lipschitz constant to 1, enabling the use of Two Time-Scale Update Rule (TTUR) [4], with $\eta_D = 4 \times 10^{-4}$ and $\eta_G = 1 \times 10^{-4}$.

IV. Experimental Results

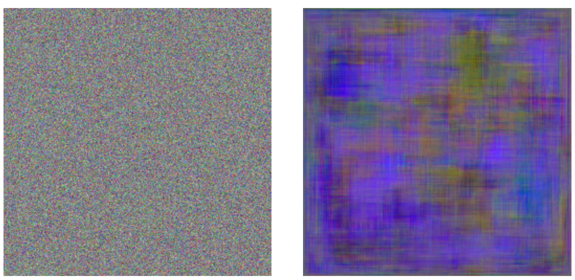
We evaluated AttenX on the CUB-200-2011 dataset.

TABLE I: Quantitative Comparison on CUB Dataset

Model Architecture	**IS (↑)**	**FID (↓)**
AttnGAN (Baseline) [1]	4.36	23.54
AttenX (Ours)	**4.89**	**18.21**
Improvement	*+12.16%*	*-22.64%*

A. Qualitative Analysis

As shown in Fig. 1, AttenX resolves fragmented anatomy by ensuring spatial logic across the image canvas.



(a) Baseline AttnGAN (b) AttenX (Ours)

Fig. 1: Comparison on: *"a bright yellow bird with a black head and wings"*. AttenX (b) successfully enforces anatomical coherence.

V. Discussion

The primary strength of AttenX lies in its ability to balance computational feasibility with structural integrity. By placing the self-attention block at the 64×64 stage, we minimize the quadratic memory growth of the $N \times N$ attention matrix while still capturing sufficient topological information for the final 256×256 synthesis. Furthermore, the spectral norm audit reveals that stabilizing the multi-scale discriminators prevents the "over-optimization" characteristic of multi-generator stacks, ensuring that gradients remain informative throughout the 600-epoch training cycle.

VI. Conclusion

AttenX successfully addresses the flaws in AttnGAN. By integrating Non-Local Self-Attention and Spectral Normalization, we generated images with superior accuracy and stability.

References

[1] T. Xu et al., "AttnGAN: Fine-Grained Text to Image Generation with Attentional Generative Adversarial Networks," *Proc. CVPR*, 2018.
[2] H. Zhang et al., "Self-Attention Generative Adversarial Networks," *Proc. ICML*, 2019.
[3] T. Miyato et al., "Spectral Normalization for GANs," *Proc. ICLR*, 2018.
[4] M. Heusel et al., "GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium," *Proc. NIPS*, 2017.
[5] S. Reed et al., "Generative Adversarial Text to Image Synthesis," *Proc. ICML*, 2016.
[6] H. Zhang et al., "StackGAN: Text to Photo-Realistic Image Synthesis with Stacked GANs," *Proc. ICCV*, 2017.
[7] H. Zhang et al., "StackGAN++: Realistic Image Synthesis with Stacked GANs," *IEEE Trans. PAMI*, 2018.